Programming Basics: Fundamentals Handbook

A structured guide to core concepts across all modern coding languages

1. Introduction to Programming

Programming is the process of writing instructions that a computer can execute to solve problems or perform specific tasks. Computers process information using binary code (0s and 1s), but humans write code in high-level programming languages (like Python, JavaScript, or C++) which are then translated into computer-readable instructions.

2. Variables and Data Types

Variables act as named containers used to store data values in a computer's memory. Every piece of data stored belongs to a specific **data type**:

Data Type	Description	Example Code
Integer	Whole numbers without decimals.	`age = 25`
Float / **Double**	Numbers containing fractional decimal points.	`price = 19.99`
String	Textual content wrapped inside quotes.	`name = "Alice"`
Boolean	Logical values representing true states.	`is_logged_in = true`

3. Control Flow Structures

Control flow determines the order in which individual statements or instructions are executed within a program.

Conditional Statements (If-Else)

Conditionals allow a program to make choices and branch execution paths based on whether a condition resolves to true or false:

```
if score >= 50:
    print("You passed!")
else:
    print("Please try again.")
```

Loops (For and While)

Loops allow block sections of code to repeat multiple times without manual duplication:

- **For Loop:** Used when the number of iterations is known in advance (e.g., iterating through a fixed list).
- **While Loop:** Continues running infinitely as long as a specified safety condition remains true.

```
# Repeats exactly 3 times
for i in range(3):
    print("Hello World")
```

4. Functions

Functions are self-contained blocks of reusable code designed to perform a distinct, single action. They eliminate code repetition and organize complex logic.

- **Parameters:** Variables accepted into the function definition as inputs.
- **Return Value:** The final output structural result sent back out from the function block to the caller.

```
def calculate_total(price, tax):
    return price + tax

# Invoking / calling the function
final_bill = calculate_total(100, 8)
```

5. Basic Data Structures

Data structures are specialized structures designed to organize, manage, and store multiple values efficiently:

1. **Arrays** / **Lists:** Ordered collections of elements where values can be accessed sequentially using an index integer starting at `0`.
2. **Dictionaries** / **Maps:** Unordered collections stored as structural `Key: Value` pairs, allowing swift data lookup based on unique keywords instead of sequence numbers.

> **Core Rule of Indexing:** In almost all programming languages, arrays are **zero-indexed**. The very first item in a list sits at position `0`, the second sits at position `1`, and so forth.